
NY HARBOR
Dx
SLUSH13
BNP
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2.1 - Chronic HF SOA

1) Floppy heart balloon:
Heart Failure = clinical syndrome resulting from impairment of ventricular filling or ejection of blood, leading to a classic set of symptoms and signs

14) Fish swimming freely:
NYHA class I = no symptoms with ordinary activity

15) Somewhat limited fish:
NYHA class II = HF symptoms with ordinary activity

16) Very limited fish:
NYHA class III = HF symptoms with less than ordinary activity

17) Resting fish:
NYHA class IV = HF symptoms at rest

30) Heart watch:
Resting sinus **tachycardia** is a marker of advanced HF **(Vital Signs)**

32) S3 **S**lushi**3**:
S3 gallop is highly specific for diagnosis of HF → best heart over apex just after S2 in early diastole**(Cardiac Auscultation)**

42) ECG grid net:
Order ECG to evaluate for underlying causes of heart failure (e.g., MI, arrhythmias) **(ECG)**

43) Treadmill next to ECG:
Every patient with suspected heart failure should undergo **stress testing** to evaluate for CAD as an underlying cause **(ECG)**

50) Heart-shaped megaphone:
Transthoracic echocardiogramis helpful in identifying important HF features **(Echo)**

Systolic Heart Failure

2) Droopy heart-shaped accordion:
Heart Failure with reduced Ejection Fraction (HFrEF) occurs primarily due to **LV SYSTOLIC dysfunction**

3) Cake slice in the shape of "<40":
Reduced EF is defined as **EF ≤40%**

18) Black heart pendant:
Ischemic heart disease (e.g., acute MI, CAD) = most common cause of HFrEF **(PMHx)**

19) Broken string on heart-shaped guitar:
Ischemic heart disease (e.g.,**acute MI**, CAD) = most common cause of HFrEF **(PMHx)**

20) Red crown with puffy yellow cap:
Ischemic heart disease (e.g., acute MI, **CAD**) = most common cause of HFrEF **(PMHx)**

21) Dilated jellyfish:
Dilated cardiomyopathy is one of the main causes of HFrEF **(PMHx)**

22) Valve breaking off saxophone:
Valvular heart disease can cause HFrEF **(PMHx)**

33) Laterally displaced heart pendant:
A **laterally displaced PMI** is indicative of LV enlargement & often found in patients with H **(Physical Exam)**

Diastolic Heart Failure

4) Diastolic diamonds:
Heart Failure with preserved Ejection Fraction (HFpEF) occurs primarily due to **LV DIASTOLIC dysfunction**

5) Candles in shape of "50":
Preserved EF is defined as **EF > 50%**

23) Steam coming out of ears:
Chronic uncontrolled **HTN** = most common cause of HFpEF **(PMHx)**

24) Harp restricting heart-shaped fish:
Restrictive cardiomyopathy (e.g., 2/2 amyloidosis, hemochromatosis, or sarcoidosis) is a common cause of impaired LV filling leading to HFpEF **(PMHx)**

25) Fish struggling to play obstructed heart-shaped bagpipes:
Hypertrophic obstructive cardiomyopathy is a common cause of HF with preserved EF → ask about a family history of sudden death at an early age (e.g., while playing sports) **(Family Hx)**

26) Shrimp overexerting on the drums:
High-output HF occurs in states of **hyperdynamic circulation** (e.g., severe chronic anemia, hyperthyroidism) or decreased SVR (e.g., AV fistula, thiamine deficiency, pregnancy)

Left Heart Failure

6) Puffed up heart-shaped pufferfish:
HF results in **increased LVEDP** → increased pressure in pulmonary circulation → increased RVEDP → increased pressure in systemic venous circulation

7) Sea turtle blowing puff of air:
Dyspnea on exertion is the most common presenting symptom of HF **(Chief Complaint)**

8) Supine sea turtle:
Orthopnea = dyspnea that occurs when lying supine, relieved by elevation of the head → highly suggestive of HF **(Chief Complaint)**

9) Sleepy angler fish:
Paroxysmal nocturnal dyspnea = awakening from sleep with shortness of breath **(Chief Complaint)**

10) Coughing sea turtle:
Many patients with left-sided HF complain of a **nonproductive cough** that is often worse at night **(Chief Complaint)**

11) Loud party blower:
Patients with HF may present with **wheezing** (aka "cardiac asthma") **(Chief Complaint)**

27) Box of rubies:
Ask about a history of cardiotoxic medications (e.g., **doxorubicin**, trastuzumab) → can cause HFrEF **(Medications)**

28) Tapestry:
Ask about a history of cardiotoxic medications (e.g., doxorubicin, **trastuzumab**) → can cause HFrEF **(Medications)**

29) Champagne bottles:
Chronic alcoholism can cause dilated cardiomyopathy & HFrEF **(Social Hx)**

30) Heart watch:
Resting sinus **tachycardia** is a marker of advanced HF **(Vital Signs)**

34) Crackly velcro:
Pulmonary congestion most commonly manifests as bibasilar **inspiratory crackles (Physical Exam)**

40) BNP blimp fish:
BNP >400 pg/mL can distinguish HF from other causes of dyspnea, but should not be used in isolation to diagnose HF **(Labs)**

41) BNP blimp fish's "400" fins:
BNP **>400** pg/mL can distinguish HF from other causes of dyspnea, but should not be used in isolation to diagnose HF **(Labs)**

44) Skull & X-bones flag:
CXR can help distinguish HF from pulmonary causes of dyspnea**(CXR)**

45) Fluffy white lung sails:
Pulmonary edema = most common CXR finding in HF, but is very nonspecific **(CXR)**

46) Large white whale:
Cardiomegaly (cardiac width >50% of thoracic width) on CXR can be suggestive of HF**(CXR)**

47) Curly sail folds:
Kerley B lines (short horizontal lines near the periphery of lungs indicative of lymphatic congestion) on CXR are suggestive of HF **(CXR)**

48) Upward-branching seaweed:
Cephalization of pulmonary vessels on CXR is suggestive of HF **(CXR)**

49) White algae-covered corners of lung sails:
Bilateral pleural effusions on CXR can be suggestive of HF **(CXR)**

51) DDx clipboard:
For further workup of Chronic HF, see Chronic Dyspnea DDx, and Acute Dyspnea DDx

Right Heart Failure

12) Bloated octopus legs:
In HF, systemic venous congestion typically presents with **peripheral edema**, most commonly in the lower extremities **(HPI)**

36) Floppy pant legs:
Systemic venous congestion in HF can manifest as **peripheral edema**, most commonly in the lower extremities **(Physical Exam)**

13) Bloated octopus abdomen:
In HF, systemic venous congestion can cause hepatomegaly and/or ascites → presenting with **abdominal distention (HPI)**

35) Big blue dilated jugs with R-shaped handles:
Elevated JVP is indicative of systemic venous congestion in HF → measure height of pulsations in the RIJ vein with patient sitting at 45 degrees **(Physical Exam)**

37) Manatee with big liver-shaped spot:
Volume overload in the splanchnic circulation can manifest as **hepatomegaly**& ascites **(Physical Exam)**

38) Yellow inner tube:
Volume overload in the splanchnic circulation can manifest as hepatomegaly & **ascites (Physical Exam)**

39) Manatee with liver-shaped spot jostling blue JVP jugs:
Hepatojugular reflux differentiates HF from non-cardiac causes of splanchnic congestion → look for sustained elevation in JVP while applying pressure to the RUQ **(Physical Exam)**